# AQ4.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Choose the set of correct options.

- [ ] If $S$ is a spanning set of the vector space $V$, then $S \cup \{v\}$ must be a spanning set of $V$, for all $v \in V$.
- [ ] Span of an empty set is the zero vector space.
- [ ] If $S$ is a spanning set of the vector space $V$, then $S \setminus \{v\}$ must be a spanning set of $V$, for all $v \in S$.
- [ ] If $S$ is a spanning set of the vector space $V$, then $S \cup \{v\}$ may not be a spanning set of $V$, for all $v \in V$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If $S$ is a spanning set of the vector space $V$, then $S \cup \{v\}$ must be a spanning set of $V$, for all $v \in V$.
Span of an empty set is the zero vector space.

Let $S$ be the set of matrices where $S = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$. Consider the following statements:

- Statement P: $Span(S)$ is the vector space consisting of only lower triangular square matrices of order 2.
- Statement Q: $Span(S)$ is the vector space consisting of only upper triangular square matrices of order 2.
- Statement R: $Span(S)$ is the vector space consisting of all the square matrices of order 2.
- Statement S: $Span(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1

*1 point*
If $S$ is a basis of $R^2$, then what is the cardinality of $S$?

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 2

*1 point*

Let $S = \{(1,2),(1,3)\}$ and $span(S) = R^n$. Then find the value of $n$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 2

*1 point*

Let $S$ be the set of matrices where $S = \left\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\right\}$. Consider the following statements:

- Statements P: $Span(S)$ is the vector space consisting of only the Identity matrix of order 2.
- Statements Q: $Span(S)$ is the vector space consisting of only diagonal matrices of order 2.
- Statements R: $Span(S)$ is the vector space consisting of all the square matrices of order 2.
- Statements S: $Span(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1

*1 point*
*1 point*

Consider the following set of vectors $S = \{(1,1,-1),(-1,1,1),(0,\frac{1}{2},0),(0,1,-2),(1,0,-2)\}$. Choose the set of correct options.

☐ The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$ ,0)} is not a basis of $R^3$.

☐ The set {(0, $\frac{1}{2}$ ,0), (0, 1, -2), (1, 0 ,-2),(1, 0, -2)} is a basis of $R^3$.

☐ The set {(-1, 1, 1), (0, $\frac{1}{2}$ ,0), (0, 1, -2)} is a basis of $R^3$.

☐ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$ ,0)} is not a basis of $R^3$.
The set {(-1, 1, 1), (0, $\frac{1}{2}$ ,0), (0, 1, -2)} is a basis of $R^3$.

*1 point*

Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in $R^2$. Choose the set of correct options.

☐ The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of $R^2$.

☐ The set { (-1, 2), (3, 1) } is a basis of $R^2$.

☐ The set { (3, 1), (-3, 1) } is a basis of $R^2$.

☐ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The set { (-1, 2), (3, 1) } is a basis of $R^2$.
The set { (3, 1), (-3, 1) } is a basis of $R^2$.

**Level 2**

*1 point*
Choose the set of correct options.

- [ ] If a subset $S$ of $R^2$ contains only two elements then $S$ is a basis of $R^2$.
- [ ] If a subset $S$ of $R^3$ contains only one elements then $S$ can never be a basis of $R^3$.
- [ ] If a subset $S$ of $R$ contains only one non zero element then $S$ is a basis of $R$.
- [ ] If $S_1$ and $S_2$ are two bases of $R^3$ then $S_1 = S_2$ .

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If a subset $S$ of $R^3$ contains only one elements then $S$ can never be a basis of $R^3$.
If a subset $S$ of $R$ contains only one non zero element then $S$ is a basis of $R$.

*1 point*
Let $V$ be the subspace of $R^3$ defined as follows:

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\}$$

Choose the set of correct options from the following.

- [ ] $\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of $V$ .
- [ ] $\{(1, 1, 0), (1, 0, -1), (0, 1, 1)\}$ is a linearly independent set of $V$ .
- [ ] $\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of $V$ .
- [ ] $\{(1, 1, 0)\}$ is a spanning set of $V$ .

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of $V$ .
$\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of $V$ .

*1 point*
Let $V$ and $W$ be vector spaces which are defined as follows:

$V = \{(x, y) \mid y = mx, \text{ where } m \neq 0 \text{ and } x, y, m \in R\}$ with usual addition and scalar multiplication as in $R^2$. $W = \{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in $R^2$.

Choose the correct set of options.

- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}$ {(1, m), (m1, 1)} is a linearly independent set in $V$ V .
- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}$ {(1, m), (m1, 1)} is a spanning set for $V$ V .
- [ ] The set $\{(1, m)\}$ {(1, m)} is a linearly independent set in $V$ V .
- [ ] The set $\{(\frac{1}{m}, 1)\}$ {(m1, 1)} is a linearly independent set in $V$ V .
- [ ] The set $\{(0, 1), (0, 2)\}$ {(0, 1), (0, 2)} is a linearly independent set in $W$ W
- [ ] The set $\{(0, 1)\}$ {(0, 1)} is a linearly independent set in $W$ W .
- [ ] The set $\{(0, 5)\}$ {(0, 5)} is a spanning set for $W$ W .

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

The set $\{(1, m), (\frac{1}{m}, 1)\}$ {(1, m), (m1, 1)} is a spanning set for $V$ V .
The set $\{(1, m)\}$ {(1, m)} is a linearly independent set in $V$ V .
The set $\{(\frac{1}{m}, 1)\}$ {(m1, 1)} is a linearly independent set in $V$ V .
The set $\{(0, 1)\}$ {(0, 1)} is a linearly independent set in $W$ W .
The set $\{(0, 5)\}$ {(0, 5)} is a spanning set for $W$ W .

Check Answers

Your score is: 0/10